ModelIt!

Teacher's Guide

Coral Reef Collapse: Death of an Underwater City

Modeling Cascading Ecosystem Failure Through Coupled Feedback Loops

Quick Reference

Grade Level	Grade 09L2
NGSS Standard	HS-LS2-6, HS-ESS3-3
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

© 2026 Discovery Collective. All rights reserved. For teacher use only.

Table of Contents

ModelIt!

1 NGSS Standards Unpacking & CAST Alignment 3

2 Scoring Rubric 4

3 Background Content Knowledge 5

4 The LEVER Framework 6

5 Slide-by-Slide Facilitation Guide 7

6 Answer Key & Expected Responses 9

7 Collaborative Learning Protocols 10

8 Differentiation & Extensions 11

9 Common Misconceptions 12

1 NGSS Standards & CAST Alignment

ModelIt!

Performance Expectation: HS-LS2-6, HS-ESS3-3

Evaluate claims, evidence, and reasoning that complex interactions in ecosystems maintain relatively consistent numbers and types of organisms. Create a computational simulation of a natural system.

Three-Dimensional Learning

Science Practice	**Using Mathematics and Computational Thinking** — Students use computational models to simulate threshold dynamics and phase shifts in a coupled ecosystem.
Disciplinary Core Idea	**LS2.C Ecosystem Dynamics, Functioning, and Resilience** — Coral reef stability depends on mutualistic relationships, trophic cascades, and structural integrity that can fail catastrophically.
Crosscutting Concept	**Stability and Change** — Students model how stable ecosystems undergo rapid irreversible phase shifts when feedback loops push past tipping points.

CAST Testing Alignment

MULTIPLE CHOICE:

Which scenario is MOST likely to cause irreversible phase shift: (a) single +2 C event, (b) three +1 C events with moderate overfishing, (c) moderate acidification alone, (d) heavy fishing alone?

CONSTRUCTED RESPONSE:

Explain why restoring a phase-shifted reef requires interventions on MULTIPLE components simultaneously.

2 Scoring Rubric

ModelIt!

Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

ModelIt!

For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Coral reefs cover less than 1 percent of the ocean but support 25 percent of all marine species. Understanding collapse requires modeling coupled feedback between thermal stress, symbiosis, trophic interactions, and ocean chemistry.

The Coral-Zooxanthellae Symbiosis

Zooxanthellae provide up to 90 percent of coral energy through photosynthesis. When thermal stress disrupts the partnership, the coral loses its energy source AND its color.

Thermal Stress Thresholds

A rise of just 1 C above summer peak for 4+ weeks triggers bleaching. If brief, recovery occurs in 6-12 months. If prolonged, coral starves and dies.

The Phase Shift Cascade

Dead coral is colonized by algae. If algae outpaces fish grazing, it smothers remaining coral and prevents larvae settlement, creating a self-reinforcing algae state.

Ocean Acidification Coupling

CO2 acidifies ocean water, weakening coral skeletons AND lowering the temperature threshold for bleaching. The two stressors are multiplicative.

4 The LEVER Framework

ModelIt!

About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L — **Locate the System**
Students identify the 7 interacting reef system components.

E — **Establish Relationships**
Students map cascading feedback loops through thermal, chemical, and biological pathways.

V — **Visualize & Model**
Students build a model showing how coupled stressors cascade to produce phase shift.

E — **Evaluate Outcomes**
Students run mild, sustained, and combined-stressor scenarios to identify thresholds.

R — **Revise & Extend**
Students add coral recruitment, sedimentation, or marine protection to explore intervention strategies.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

ModelIt!

Slide 1: Cover

2 min

Say: *"This reef took 10,000 years to grow. It died in 10 years."*

Watch For

Show before/after images.

Slide 2: Learning Objectives

3 min

Say: *"Today we model the coupled feedback loops that make reefs both productive and fragile."*

Watch For

Pre-teach zooxanthellae and phase shift.

Slide 3: Big Question

3 min

Say: *"Your body at 37 C is fine. At 40 C your organs fail. Corals are the same."*

Watch For

Quick-write: How many connections can you identify?

Slide 4: LEVER Framework

2 min

Say: *"Seven components with cascading connections."*

Watch For

Preview coupled biological + chemical + structural pathways.

Slide 5: Activity 1: Components

8 min

Say: *"Which can humans directly control?"*

Watch For

Temperature and acidity are external; rest are ecological responses.

Slide 6: Activity 2: Connect

10 min

Say: *"Two pathways to destruction -- thermal and chemical -- and they INTERACT."*

Watch For

Map both pathways. Show how acidification lowers bleaching threshold.

Slide 7: Activity 3: Simulate

10 min

Say: *"Is one big problem worse than three small ones?"*

Watch For

Students discover combined stressors exceed single severe stressor.

Slide 8: Discoveries

5 min

Say: *"Once the phase shift happens, there is no going back."*

Watch For

Discuss irreversibility and climate urgency.

Slide 9: STEM Challenge

12 min

Say: *"You are chief scientist for the Great Barrier Reef. What is your plan?"*

Watch For

Groups design monitoring and intervention systems.

6 Answer Key & Expected Responses

ModelIt!

Component Sorting

- **EXTERNAL:** Ocean Temperature, Water Acidity
- **INTERNAL:** Coral Bleaching Rate, Algae Overgrowth, Fish Population, Symbiotic Zooxanthellae, Reef Structural Integrity

Relationships

- **Ocean Temperature to Bleaching Rate:** POSITIVE (+) — Higher temperatures trigger thermal stress and bleaching.
- **Bleaching Rate to Zooxanthellae:** NEGATIVE (-) — Bleaching expels zooxanthellae from coral tissue.
- **Zooxanthellae to Structural Integrity:** POSITIVE (+) — Healthy symbionts provide energy for coral growth and skeleton maintenance.
- **Structural Integrity to Fish Population:** POSITIVE (+) — Reef structure provides habitat for herbivorous fish.
- **Fish Population to Algae Overgrowth:** NEGATIVE (-) — Herbivorous fish graze algae, preventing smothering.
- **Algae Overgrowth to Bleaching Rate:** POSITIVE (+) — Algae smother coral, adding stress that accelerates death.
- **Water Acidity to Structural Integrity:** NEGATIVE (-) — Acidification weakens coral skeletons.
- **Water Acidity to Bleaching Rate:** POSITIVE (+) — Acidification lowers the temperature threshold for bleaching.

Key Discoveries

- Coral bleaching creates a cascading feedback loop from heat through starvation to permanent phase shift
- There is a temperature threshold beyond which bleaching accelerates faster than recovery
- Multiple moderate stressors can trigger collapse when no single stressor alone would be sufficient
- The phase shift to algae dominance represents a stable alternative state that persists even after stress removal

7 Collaborative Learning Protocols

ModelIt!

Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

ModelIt!

Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

ModelIt!

Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Bleaching means coral is dead"	Bleaching is stress, not death. Coral is alive but starving. If stress is removed within weeks, zooxanthellae recolonize. Death occurs with prolonged bleaching.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"If we stop warming reefs will recover"	Stopping warming is necessary but not sufficient after a phase shift. Algae maintain dominance through their own feedback loops. Recovery requires active multi-component intervention.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Reefs will migrate to cooler water"	Reef migration is not feasible. Reefs need specific conditions and centuries to build structure. Warming rates are 50-100x faster than natural migration.